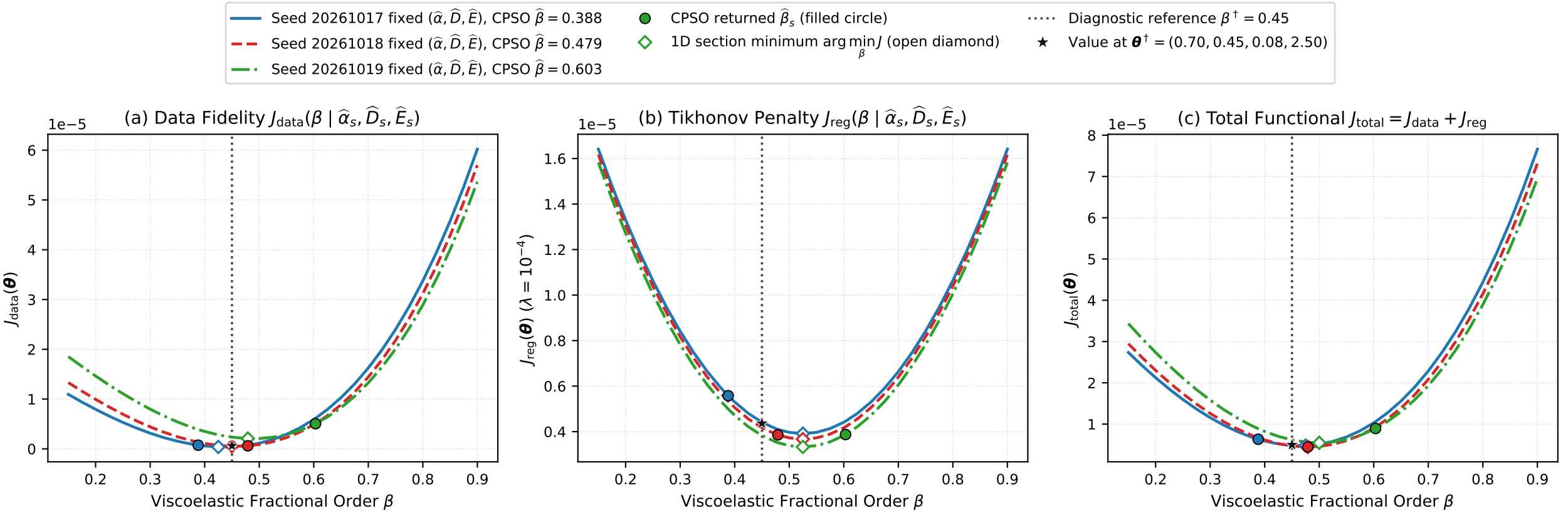
Seed 20261017 fixed $(\widehat{\alpha}, \widehat{D}, \widehat{E})$, CPSO $\widehat{\beta} = 0.388$
Seed 20261018 fixed $(\widehat{\alpha}, \widehat{D}, \widehat{E})$, CPSO $\widehat{\beta} = 0.479$
Seed 20261019 fixed $(\widehat{\alpha}, \widehat{D}, \widehat{E})$, CPSO $\widehat{\beta} = 0.603$
CPSO returned $\widehat{\beta}_s$ (filled circle)
1D section minimum $\arg\min_{\beta} J$ (open diamond)
Diagnostic reference $\beta^{\dagger} = 0.45$
Value at $\boldsymbol{\theta}^{\dagger} = (0.70, 0.45, 0.08, 2.50)$
(a) Data Fidelity $J_{\text{data}}(\beta \mid \widehat{\alpha}_s, \widehat{D}_s, \widehat{E}_s)$
1e−5
$J_{\text{data}}(\boldsymbol{\theta})$
Viscoelastic Fractional Order β
(b) Tikhonov Penalty $J_{\text{reg}}(\beta \mid \widehat{\alpha}_s, \widehat{D}_s, \widehat{E}_s)$
1e−5
$J_{\text{reg}}(\boldsymbol{\theta})$ $(\lambda = 10^{-4})$
Viscoelastic Fractional Order β
(c) Total Functional $J_{\text{total}} = J_{\text{data}} + J_{\text{reg}}$
1e−5
$J_{\text{total}}(\boldsymbol{\theta})$
Viscoelastic Fractional Order β